

Part A · Multiple Choice

2 points each

1. **c) That summer is warm because the Sun is physically closer to Earth**
p. 70 — the real cause is the 23.5° axial tilt combined with Earth's revolution, which changes how directly the Sun's rays strike each hemisphere.
2. **c) UVC**
p. 69 — ozone removes UVC completely, most UVB, and some UVA. UVA, the lowest energy, reaches the ground.
3. **d) Chlorofluorocarbons (CFCs)**
p. 70 — CFCs are the single exception on the list; the text notes they have only recently existed.
4. **a) Stratosphere**
p. 81 — between roughly 15 and 30 km altitude. This is why a traveler in the mesosphere, which sits above it, would receive severe UV burns.
5. **c) Warm air sits beneath cooler air, making it unstable and causing constant mixing**
p. 80 — that instability and mixing is precisely why all of the planet's weather happens in the troposphere.
6. **a) The flame has a higher temperature but less heat**
pp. 84–85 — temperature measures how fast atoms vibrate; heat measures total energy. The bathtub is cooler but holds far more vibrating atoms.
7. **d) from the west toward the east**
p. 93 — wind belts are named for the direction winds come from. This is also why a flight from San Francisco to New York is faster than the return trip.
8. **b) the Ferrel and Polar cells**
p. 93 — relatively warm, moist Ferrel air meets cold, dry Polar air, making the weather there extremely variable across much of North America and Europe.
9. **a) temperature stays the same but humidity increases, or humidity stays the same but temperature decreases**
pp. 97–98 — water vapor then condenses around a nucleus such as dust, smoke, or a salt crystal, and billions of these droplets together make a cloud.
10. **d) continental polar (cP)**
p. 101 — “continental” means it formed over land and is dry; “polar” means it formed at high latitudes and is cold.
11. **b) entrainment**
p. 106 — entrainment happens once precipitation becomes too heavy for the updrafts to hold, marking the transition into the mature stage.
12. **b) Dry (B)**
p. 111 — note that the Dry classification is about the water balance, not about temperature.

Part B · Matching

8 points each

13. **1–d · 2–f · 3–b · 4–a · 5–h · 6–e · 7–g · 8–c**
1 point each. Advection p. 85 · Latent heat p. 84 · Albedo p. 83 · Dew point p. 96 · Source region p. 101 · Squall line p. 102 · Nor'easter p. 108 · Eyewall p. 109.

14. In the troposphere temperature decreases with altitude, so warm air lies beneath cool air — an unstable arrangement that overturns. In the stratosphere temperature increases with altitude, so warm air lies above cool air, which is stable.

Rubric: 1.5 pts for the troposphere — warmer, less dense air beneath cooler air is buoyant and rises while cool air sinks, producing constant mixing (and all of Earth's weather). 1.5 pts for the stratosphere — warmer, less dense air already sits on top, so nothing rises and there is little mixing. Credit consequences as supporting evidence: volcanic ash stays suspended in the stratosphere for years, and pilots fly its lower portion for smooth air. (pp. 80–81)

15. Moist air forced up the windward side of a mountain range cools and drops its precipitation there; the now-dry air descends the leeward side, warming and drying the land below.

Rubric: 1 pt air rises over the windward side, cools, and forms clouds and precipitation. 1 pt the air that sinks on the leeward side is already dry, and descending air warms and creates strong dry winds. 1 pt the result is a persistently dry leeward zone — the text credits this effect with creating many of the world's deserts. Credit Chinook winds (which can raise temperatures more than 20°C in an hour) as the named example. (p. 89)

16. Cumulus, mature, and dissipating.

Rubric: 1 pt each. **Cumulus** — an air parcel rises, cools, and condenses; latent heat released by condensation warms the cloud and drives it upward, and updrafts hold precipitation aloft. **Mature** — precipitation grows too heavy for the updrafts and begins to fall, creating downdrafts; updrafts and downdrafts exist side by side, an anvil may form at the top of the troposphere, and the storm is at its most intense. **Dissipating** — downdrafts overtake the updrafts, cutting off the latent heat supply, and the storm weakens into light precipitation. (p. 106)

17. It supplies the gases living things need and cycles them through photosynthesis and respiration, and its greenhouse gases moderate Earth's temperatures.

Rubric: 1.5 pts each for any two of the following. **Gases for life** — plants use CO₂ and produce O₂ in photosynthesis, which accounts for nearly all atmospheric oxygen; animals use that oxygen in respiration. **Temperature moderation** — greenhouse gases trap heat, so temperatures do not swing between frigid nights and scorching days as they would on an airless world like the Moon. **UV protection** — the ozone layer absorbs most harmful ultraviolet radiation, which would otherwise penetrate cells and damage DNA. (pp. 77 and 81)

18. See rubric.

Rubric: 1.5 pts — unequal heating. More solar energy strikes the equator because the Sun's rays arrive nearly straight on there, while the same energy is spread over a larger area at higher latitudes. Warm tropical seas reach 28°C (82°F) or higher in summer and autumn.

1.5 pts — evaporation and latent heat. The warm sea evaporates water, creating a large humid air mass. As that air rises and its water vapor condenses, latent heat is released, warming the air further and driving it higher still — a self-feeding engine. The text notes a single hurricane can release energy approaching the total annual electrical consumption of the United States.

1.5 pts — low pressure. Rising warm air leaves a low-pressure zone at the surface (a tropical depression). Surrounding air rushes in to fill it, feeding still more warm moist air into the system, and thunderstorms build around it.

1.5 pts — Coriolis effect and structure. Earth's rotation deflects the inrushing air, setting the storm rotating counterclockwise in the Northern Hemisphere and clockwise in the Southern. Given low wind shear, the system organizes into a hurricane within two to three days, with a calm eye, an eyewall of strongest winds, and spiral rain bands.

Full credit for a clear causal chain even if the wording differs. Credit the closing observation that a hurricane weakens over cooler water or land because its latent heat source is cut off. (pp. 69–70, 84, 90–91, 108–109)